AI Coding Assistant Brief

v1 — Python Hybrid Analyzer

- Validates C++ syntax using `g++`.
- Runs static rules to detect common bug/security patterns.
- First it compiles the code to catch syntax errors, using g++. If compilation fails, it captures the error output for analysis.
- Then applies regex-based rules for deeper analysis, particularly for runtime errors and common coding mistakes.
- Finally calls LLM (Gemini/Ollama - qwen3:8b) to generate explanations and guided fixes.

Output Images

Explanation using Gemini:

```
INFO:     127.0.0.1:65366 - "POST /analyze HTTP/1.1" 200 OK
2026-02-18 20:09:29,998 [INFO] [203fcc4f-90ac-4421-94f6-f9515bb4e1c2] Processing bundle test-dsa-explain
2026-02-18 20:09:30,200 [INFO] Compilation completed in 0.201s
2026-02-18 20:09:30,201 [INFO] [203fcc4f-90ac-4421-94f6-f9515bb4e1c2] Syntax validation time: 0.20 seconds
2026-02-18 20:09:30,202 [INFO] [203fcc4f-90ac-4421-94f6-f9515bb4e1c2] Total Analysis time: 0.20 seconds
2026-02-18 20:09:52,851 [INFO] HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent?key=AIzaSyDjN-93pfQbvTz
UaJ7I--iQv7JVxwvscRA "HTTP/1.1 200 OK"
2026-02-18 20:09:52,853 [INFO] [test-dsa-explain] Explanation time: 22.65 seconds
INFO:     127.0.0.1:65367 - "POST /analyze/explain HTTP/1.1" 200 OK
```

Explanation using Ollama:

```
INFO:     127.0.0.1:59620 - "POST /analyze HTTP/1.1" 200 OK
2026-02-18 21:36:29,507 [INFO] [d176acae-3c0a-42c9-8579-c3151b47161e] Processing bundle test-dsa-explain
2026-02-18 21:36:29,701 [INFO] Compilation completed in 0.194s
2026-02-18 21:36:29,701 [INFO] [d176acae-3c0a-42c9-8579-c3151b47161e] Syntax validation time: 0.19 seconds
2026-02-18 21:36:29,702 [INFO] [d176acae-3c0a-42c9-8579-c3151b47161e] Total Analysis time: 0.20 seconds
2026-02-18 21:37:32,756 [INFO] HTTP Request: POST http://localhost:11434/api/generate "HTTP/1.1 200 OK"
2026-02-18 21:37:32,759 [INFO] [test-dsa-explain] Explanation time: 63.06 seconds
INFO:     127.0.0.1:59622 - "POST /analyze/explain HTTP/1.1" 200 OK
```

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v2 — Node.js Multi-Stage LLM Pipeline

- Accepts code and numbers lines for better issue referencing.
- Runs 3 LLM stages: **Detect** → **Validate** → **Explain**.
- Currently using same Ollama Qwen 3.8b model for all stages, but designed to allow different models per stage.
- Having different system prompts for each stage to optimize for the specific task (detection, validation, explanation).
- Enforces JSON structure with Zod and retries invalid model output.

Output Images

```
# intelligence-enginev2 > node .\test-run.js
[2026-02-18T16:48:00.877Z] [DEBUG] Sending prompt to model {"model":"qwen3:8b","endpoint":"http://localh
ost:11434/api/generate"}
[2026-02-18T16:48:45.767Z] [DEBUG] Attempt 1/2 — raw model output received
[2026-02-18T16:48:45.769Z] [INFO] ⏱  Detect completed {"duration":"44.891901700 s"}
[2026-02-18T16:48:45.769Z] [DEBUG] Detection results {"findingCount":5}
[2026-02-18T16:48:45.770Z] [DEBUG] Sending prompt to model {"model":"qwen3:8b","endpoint":"http://localh
ost:11434/api/generate"}
[2026-02-18T16:49:21.968Z] [DEBUG] Attempt 1/2 — raw model output received
[2026-02-18T16:49:21.969Z] [INFO] ⏱  Validate completed {"duration":"36.199480900 s"}
[2026-02-18T16:49:21.970Z] [DEBUG] Validation results {"findingCount":5}
[2026-02-18T16:49:21.970Z] [DEBUG] Sending prompt to model {"model":"qwen3:8b","endpoint":"http://localh
ost:11434/api/generate"}
[2026-02-18T16:50:26.358Z] [DEBUG] Attempt 1/2 — raw model output received
[2026-02-18T16:50:26.363Z] [INFO] ⏱  Explain completed {"duration":"64.392897800 s"}
[2026-02-18T16:50:26.363Z] [DEBUG] Explanation results {"findingCount":5}
[2026-02-18T16:50:26.364Z] [INFO] ✅ Analysis complete {"totalDuration":"145.486951000 s","findings":5}
```


```
=== ANALYSIS RESULT ===
{
  "findings": [
    {
      "line_range": [
        6,
        8
      ],
      "category": "bug",
      "severity": "major",
      "issue": "Off-by-one error in array indexing",
      "why_it_matters": "Initializing 'right' to arr.size() accesses memory beyond the array bounds, lea
ding to undefined behavior or crashes",
      "hint": "Array indices should be 0-based with maximum value size()-1",
      "guided_fix": "Change 'right = arr.size()' to 'right = arr.size() - 1'",
      "confidence": 1
    },
    {
      "line_range": [
        23,
        23
      ],
      "category": "bug",
      "severity": "major",
      "issue": "Incorrect loop termination condition",
      "why_it_matters": "The condition 'i <= n' allows the loop to execute when i equals n, accessing el
ements beyond the array's valid range",
      "hint": "Bubble sort should iterate while i < n to prevent out-of-bounds access",
      "guided_fix": "Modify the loop condition to 'while (i < n)'",
      "confidence": 1
    },
```

```
    {
      "line_range": [
        33,
        33
      ],
      "category": "security",
      "severity": "critical",
      "issue": "Potential division by zero vulnerability",
      "why_it_matters": "Dividing by a variable that can be zero creates a runtime exception and potenti
al security exploitation vectors",
      "hint": "Always validate denominators before performing division operations",
      "guided_fix": "Add a check: if (denominator != 0) { ... } else { handle error }",
      "confidence": 1
    },
    {
      "line_range": [
        37,
        38
      ],
      "category": "bug",
      "severity": "major",
      "issue": "Invalid vector element access",
      "why_it_matters": "Accessing index 0 of an empty vector triggers undefined behavior and potential
program termination",
      "hint": "Check vector emptiness before accessing elements",
      "guided_fix": "Add 'if (!vec.empty()) { ... } else { handle empty case }",
      "confidence": 1
    },
```

```
    {
      "line_range": [
        40,
        40
      ],
      "category": "bug",
      "severity": "major",
      "issue": "Assignment vs equality comparison error",
      "why_it_matters": "The assignment operator '=' creates a side effect that alters program logic ins
tead of performing a comparison",
      "hint": "Use '==' for equality checks and '=' for assignments",
      "guided_fix": "Change 'if (x=5)' to 'if (x==5)'",
      "confidence": 1
    }
  ]
}
```


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v3 — Node.js MVC Gemini Engine

- Uses Express 5 MVC + middleware for production-grade request handling.
- Sends a language-aware single-pass prompt to Gemini.
- Normalizes/cleans model response before returning API output.

Output Images

```
# intelligence-enginev3 > node .\scripts\test-api.mjs

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║        Intelligence Engine v3 — API Smoke Test       ╖

  Target: http://localhost:3000/code_review
  Time:   2026-02-18T17:10:29.322Z

── Response ──────────────────────────────────
  Status:     200 OK
  Request-Id: 8123be56-bc07-4383-bdf0-0674681edc30
  Risk Level: high
  Quality:    0
  Findings:   3

── Full Response ─────────────────────────────
{
  "summary": {
    "risk_level": "high",
    "overall_quality": 0
  },
  "findings": [
    {
      "category": "bug",
      "severity": "critical",
      "line_range": [
        25,
        25
```

— Full Response ————————————————————————


```
{
  "summary": {
    "risk_level": "high",
    "overall_quality": 0
  },
  "findings": [
    {
      "category": "bug",
      "severity": "critical",
      "line_range": [
        25,
        25
      ],
      "issue": "Attempting to access `v[0]` on an empty `std::vector<int> v`.",
      "why_it_matters": "This is an out-of-bounds access, leading to undefined behavior and a high pr
obability of a program crash at runtime. Accessing elements of an empty vector via `operator[]` is ne
ver safe.",
      "hint": "Before accessing elements, ensure the vector is not empty or has been properly sized a
nd populated.",
      "guided_fix": "Initialize the vector with elements, resize it, or check `v.empty()` before atte
mpting to access `v[0]`.",
      "full_fix": null
    },
    {
      "category": "bug",
      "severity": "critical",
      "line_range": [
        20,
        20
      ],
      "issue": "The `divide` function performs integer division `a / b` without validating that the d
ivisor `b` is non-zero.",
      "why_it_matters": "If `b` is zero (as demonstrated by the call on line 30), this operation resu
lts in a division-by-zero error, which is a critical runtime exception and will crash the program.",
      "hint": "Always validate function parameters that could lead to critical errors like division b
y zero.",
      "guided_fix": "Add a check for `b == 0` at the beginning of the `divide` function. Handle this
case appropriately, for example, by throwing an exception, returning an error code, or printing an er
ror message and exiting.",
```

```
      "guided_fix": "Add a check for `b == 0` at the beginning of the `divide` function. Handle this
case appropriately, for example, by throwing an exception, returning an error code, or printing an er
ror message and exiting.",
      "full_fix": null
    },
    {
      "category": "bug",
      "severity": "critical",
      "line_range": [
        7,
        7
      ],
      "issue": "In the `binarySearch` function, the `right` index is initialized to `arr.size()`.",
      "why_it_matters": "For an inclusive search range `[left, right]`, `right` should be initialized
 to `arr.size() - 1`. Initializing `right` to `arr.size()` can lead to `mid` potentially evaluating t
o `arr.size()` during the search (e.g., when `left` and `right` both become `arr.size()`), causing an
 out-of-bounds access when `arr[mid]` is evaluated. This results in undefined behavior and a potentia
l program crash.",
      "hint": "Carefully define the search range (inclusive or exclusive) and ensure all index calcul
ations and loop conditions adhere to it to prevent out-of-bounds access.",
      "guided_fix": "Change `int right = arr.size();` to `int right = arr.size() - 1;` to correctly d
efine the inclusive upper bound for the search range.",
      "full_fix": null
    }
  ]
}

── Timing Report (highest precision) ──────────────
  Nanoseconds:  18244397900 ns
  Milliseconds: 18244.3979 ms
  Seconds:      18.2443979 sec

── Result ─────────────────────────────────────────
  ✅ PASS — API returned a successful response.
# intelligence-enginev3 >
```

Github Repository (dev - branch)

- https://github.com/DeepPatel4505/MentiCode-Backend.git